

PROJECT DESIGN PHASE: SOLUTION ARCHITECTURE

WeatherWise AI — System Architecture & Component Design

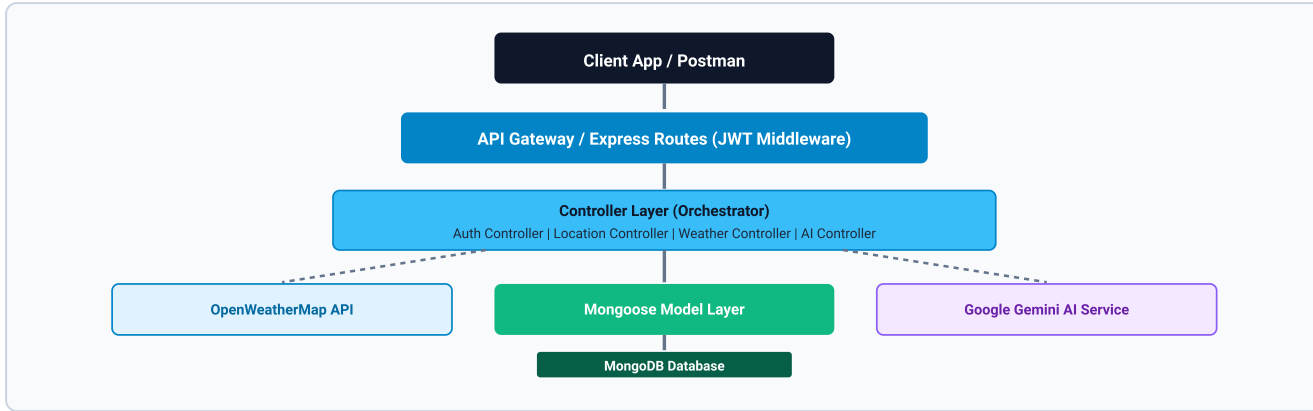
DATE
29/09/26

TEAM ID
SWTID-2026-8956

PROJECT NAME
WeatherWise AI

MAX MARKS
4 Marks

1. SYSTEM ARCHITECTURE DIAGRAM (MVC PATTERN)



Technical Stack

- **Runtime & Framework:** Node.js (v16+) & Express.js
- **Database & ODM:** MongoDB & Mongoose ODM
- **AI Engine:** Google Gemini AI SDK (@google/generative-ai)
- **Security & Auth:** JWT & bcryptjs password hashing
- **Testing Tools:** Postman / Thunder Client

Hardware & Environment

- **OS Support:** Windows 10/11, macOS, Linux
- **Processor:** Intel Core i5 (8th Gen+) / AMD Ryzen 5
- **Memory (RAM):** 8 GB Minimum (16 GB Recommended)
- **Storage:** 1 GB available workspace

2. ARCHITECTURAL HIGHLIGHTS & SECURITY

- **Authentication Middleware:** Intercepts secured endpoints to validate JWT signatures before routing request execution to controllers.
- **Resilient Fallback Mode:** Maintains basic meteorological API responses even if external AI/weather API keys are offline or rate-limited.
- **Sanitization & Security:** Built-in request parameter sanitization guarding against XSS and NoSQL injection vulnerabilities.

3. DATABASE SCHEMAS & ENTITY MAPPINGS

MongoDB database collections managed seamlessly via Mongoose Object Data Modeling (ODM):

1. User Schema	Collection: users	2. Location Schema	Collection: locations
<code>_id</code>: ObjectId (Primary Key) <code>name</code>: String (Required) <code>email</code>: String (Required, Unique, Lowercase, Regex validated) <code>password</code>: String (Hashed with bcryptjs, min 6 chars) <code>timestamps</code>: Automatically manages createdAt & updatedAt		<code>_id</code>: ObjectId (Primary Key) <code>city</code>: String (Required) <code>country</code>: String (Required) <code>user</code>: Foreign Key referencing User Collection Constraint: Compound index { user: 1, city: 1, country: 1 } prevents duplicate locations per user.	

4. USER ROLES & ACCESS HIERARCHY

Role	Permissions & Capabilities
Admin	Manages global user accounts, oversees database integrity, and monitors API health/server logs.
Registered User	Manages favorite locations (CRUD), requests live metrics, and receives AI recommendations via Gemini.
Public User	Interfaces with open public endpoints to view raw weather data without requiring account creation.

5. END-TO-END USER EXECUTION WORKFLOW

1 Authentication Register/Login at /api/auth to receive JWT token.	2 Save Location Post favorite city via /api/locations route.	3 Fetch Metrics Query live weather via /api/weather/:city.	4 AI Query Request insights via /api/ai/recommendation.	5 JSON Response Receive tailored clothing & activity recommendations.
--	--	--	---	---

Team Responsibility Sign-off: Designed and validated by Team SWTID-2026-8956 (Sanmathi Kumar, Arifa Zohra, Ainul Liyana P M, Vaishali S) as part of the WeatherWise AI Project Design Phase submission.